HANOI UNI. OF SCI. AND TECH. **FACULTY OF MATH. AND INFORMATICS**

MIDTERM TEST, SPRING 2026 Course: CS370 - Nat. of PL Duration: 60 minutes	Student name: ..... Student Id: ..... Order: ... Class Id: .....

Evaluators' name and signature	Proctors' name and signature	Total score

Test Id: 3 **(This test consists of 10 questions)**

Note: *No electronic devices (laptops, smart devices, etc. except calculators) are allowed.*

Questions with one correct answer

Question 1. In Prolog, suppose that the following clauses

$$ancestor(X, Y) :- parent(X, Z),\ ancestor(Z, Y).$$
$$ancestor(amy, bob).$$
$$parent(eve, amy).$$

have been entered into the database, then which of the following queries would cause the "yes" response?

- ☐ `?- ancestor(amy,eve).`
- ☐ `?- ancestor(bob,amy).`
- ☐ `?- ancestor(eve,bob).`
- ☐ `?- ancestor(eve,amy).`

Question 2. Which of the following Pascal programs causes a syntax error?

- ☐ `var i: Integer; begin Writeln(i=1); end.`
- ☐ `var i: Integer; begin Writeln; i:=1.0; end.`
- ☐ `var i: Integer; begin Writeln; i:=1; end.`
- ☐ `var i: Integer; begin Writeln(i:=1); end.`

Question 3. In Ada, which of the following statements is legal?

- ☐ `I = I * 1;`
- ☐ `while I >= 10 do J := J - 1; end while;`
- ☐ `while J <= 10 loop J := J + 1; end loop;`
- ☐ `while I <= 5 do J := J + 1; end loop;`

Question 4. In Java, which of the following statements is legal?

- ☐ `int v; if (v == 0) v=v+1;`
- ☐ `System.out.println("Hello World");`
- ☐ `System.out.println(Hello World);`
- ☐ `System.out.println('Hello World');`

Question 5. In C#, which of the following statements is legal?

- ☐ `if (int v == 0) return v;`
- ☐ `int v; if (v == 0) v=v+1;`
- ☐ `Console.WriteLine("Hello World");`
- ☐ `Console.out.println("Hello World");`

Question 6. In Python, which of the following statements is legal?

- ☐ `printf ('Hello World');`
- ☐ `if (v != 0) v=v+1;`
- ☐ `sum := float(1) + float(2);`
- ☐ `println('Hello World');`
- ☐ `print ("Hello World");`
- ☐ `Console.out.println("Hello World");`

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Constructed-response questions

Question 7. Describe briefly your role and tasks in your group project (i.e. the project that your group has taken in this course).

Question 8. Explain shortly why C++ is faster than Java (i.e. 1.0 vs 1.7 for speed).

Question 9. Give an imperative form in C of the following Pascal code.

```
function prod (n: Integer): Real;
  begin
    if n>=1 then
      prod:=n*prod(n-1)
    else
      prod:=1.0;
  end
```

Question 10. Give a piece of code in C++, that is equivalent to the following code in Java.

```
class A
{ A A(A A)
  { A:
    for(;;)
    { if (A.A(A) == A) break A; }
    return A;
  }
}
```

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Evaluators' name and signature	Proctors' name and signature	Total score

Test Id: 4 **(This test consists of 10 questions)**

Note: *No electronic devices (laptops, smart devices, etc. except calculators) are allowed.*

Questions with one correct answer

Question 1. In Prolog, suppose that the following clauses

$$ancestor(X, Y) :- parent(X, Z),\ ancestor(Z, Y).$$
$$ancestor(amy, bob).$$
$$parent(eve, amy).$$

have been entered into the database, then which of the following queries would cause the "no" response?

- ☐ `?- parent(eve,amy).`
- ☐ `?- ancestor(amy,bob).`
- ☐ `?- ancestor(eve,bob).`
- ☐ `?- ancestor(bob,eve).`

Question 2. Which of the following Pascal programs causes an incompatible type error?

- ☐ `var i: Integer; begin Writeln(i:=1); end.`
- ☐ `var i: Integer; begin Writeln(i=1); end.`
- ☐ `var i: Integer; begin Writeln; i:=1.0; end.`
- ☐ `var i: Integer; begin Writeln; i:=1; end.`

Question 3. In Ada, which of the following statements is legal?

- ☐ `while I >= 10 do J := J - 1; end while;`
- ☐ `while J <= 10 loop J := J + 1; end loop;`
- ☐ `while I <= 5 do J := J + 1; end loop;`
- ☐ `I = I * 1;`

Question 4. In Java, which of the following statements is legal?

- ☐ `if (int v = 0) return v;`
- ☐ `System.out.println('Hello World');`
- ☐ `int v; if (v == 0) v=v+1;`
- ☐ `int v=1; if (v != 0) v=v+1;`

Question 5. In C#, which of the following statements is legal?

- ☐ `System.out.println("Hello World");`
- ☐ `if (int v == 0) return v;`
- ☐ `Console.WriteLine("Hello World");`
- ☐ `int v; if (v != 0) v=v+1;`

Question 6. In Python, which of the following statements is legal?

- ☐ `if (v != 0) v=v+1;`
- ☐ `sum = float(1.0) + float(2);`
- ☐ `printf ("Hello World");`
- ☐ `sum := float(1) + float(2);`
- ☐ `Console.println("Hello World");`
- ☐ `println('Hello World');`

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Constructed-response questions

Question 7. Describe briefly your contribution to your group project (i.e. the project that your group has taken in this course).

Question 8. Explain shortly why C++ is more memory-efficient than Java (i.e. 1.6 vs 9.1 for space).

Question 9. Give a functional form in C of the following Pascal code.

```
function sum (n: Integer): Integer;
  var i,s: Integer;
  begin
    s:=0;
    if n<=0 then
      sum:=0;
    else
      for i := 1 to n do
        s:=s+i;
    sum:=s;
  end
```

Question 10. Give a piece of code in Java, that is equivalent to the following code in C++.

```
enum Disc {IsInt,IsReal};
struct IntOrReal {
  enum Disc which;
  union {
    int i;
    double r;
  };
};
```